ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(a)** 28	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3	**(b)** -4	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
4	**(b)** 29	$(\alpha+\beta)^2 - 2\alpha\beta$
5	**(d)** 0	Common difference is -1; $a_{p+q} = 0$
6	**(d)** 8	$(m+n)^2 - m^2$
7	**(a)** 12	Let the point be $(0, y)$
8	**(b)** $\frac{25}{9}$	$1 + \tan^2\theta$
9	**(c)** $2+2\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3}-1)$; Height $= 2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m)
10	**(d)** 48 cm	$AM = \frac{30\times40}{50} = 24$ cm; $AB = 2AM = 48$ cm
11	**(c)** 4.47 cm²	Sector $= 25.67$ cm²; triangle $= \frac{\sqrt{3}}{4}r^2 = 21.19$ cm²; Segment $= 25.67 - 21.19 = 4.47$ cm²
12	**(a)** 4410 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³
13	**(d)** 23	$\frac{4325+25p}{77+p} = 49$ gives $(25-49)p = -552$; $p = 23$
14	**(d)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
15	**(b)** 60	LCM of 1 to 6
16	**(b)** 30	$\alpha\beta(\alpha+\beta)$
17	**(c)** 2	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
18	**(c)** 24	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
19	**(b)** 54	$a_n = \frac{2S_n}{n} - a$
20	**(c)** 6	Find b by Pythagoras

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Numbers are $12 \times 3 = 36$ and $12 \times 5 = 60$ (1); LCM $= 180$ (1).	1+1
22	New sum $= \alpha^2+\beta^2 = 10$ (1); product $= (\alpha\beta)^2 = 9$; polynomial $x^2 - 10x + 9$ (1).	1+1
23	$\frac{k}{8} = \frac{2}{k} \Rightarrow k = \pm 4$ (1); $k = 4$ gives coincident lines, so $k = -4$ (1).	1+1
24	$D = (4)^2 - 4k(1) = 0$ (1); $k = 4$ (1).	1+1
25	$a+3d = 9$, $a+8d = 4$ (1); $d = -1$, $a = 12$ so $a_{13} = 0$ (1).	1+1
26	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{7}$ (1); $DE = 6$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	LCM$(18, 24, 36) = 72$ min (2); rounds 4, 3, 2 (1).	2+1
28	$\alpha+\beta = 3$, $\alpha\beta = 2$ (1); sum $= (\alpha+\beta) + \frac{\alpha+\beta}{\alpha\beta} = \frac{9}{2}$ (1); product $= \alpha\beta + 2 + \frac{1}{\alpha\beta} = \frac{9}{2}$ (1).	1+1+1
29	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $2p+3q = 2$, $3p+5q = \frac{19}{6}$ (1); eliminating q: $(1)p = \frac{1}{2}$ so $p = \frac{1}{2}$ (1); $q = \frac{1}{3}$ (1); $x = 2$, $y = 3$ (1).	1+1+1+1
30	LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $3(x^2+1) = 5(x^2-1)$ (1); $x^2 = 4$ so $x = \pm 2$ (1).	1+1+1
31	$\frac{6}{2}(2a+5d) = 75$ and $\frac{10}{2}(2a+9d) = 185$ (1); simplify to linear equations in a, d (1); $a = 5$, $d = 3$, AP $5, 8, 11, 14, \ldots$ (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 2:3$ (1); area ratio $= 4:9$ (1).	1+1+1

32	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
	OR: LHS $= \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$ (2); $= \dfrac{1}{\sin\theta\cos\theta} = \sec\theta\csc\theta$ (1).	2+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Books per stack $=$ HCF$(96, 240, 336) = 48$ (3); stacks $= \dfrac{672}{48} = 14$ (2).	3+2
34	Let zeroes be $a-d, a, a+d$; sum $= 3a = 6$ so $a = 2$ (2); product $= a(a^2 - d^2) = 6$ gives $d^2 = 1$ (2); zeroes $1, 2, 3$ (1).	2+2+1
35	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $36p + 40q = 10$, $24p + 70q = 11$ (2); $p = \frac{1}{6}$, $q = \frac{1}{10}$ (1); $x - y = 6$, $x + y = 10$ (1); boat 8 km/h, stream 2 km/h (1).	2+1+1+1
36	Smaller tap x h, larger $x - 5$ h: $\dfrac{1}{x} + \dfrac{1}{x-5} = \dfrac{1}{\frac{10}{3}}$ (2); simplify to a quadratic and solve, rejecting the root less than 5: $x = 10$ (2); times 10 h and 5 h (1).	2+2+1
	OR: AP with $a = 100$, $l = 40$, $n = 21$ (1); $S_n = \frac{21}{2}(100 + 40) = 1470$ cm (2); middle term $= \frac{100+40}{2} = 70$ cm (2).	1+2+2
37	Right triangle with legs 60 and 80 (1); distance $= \sqrt{3600 + 6400} = 100$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Centroid $= (6, \frac{8}{3})$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 48$ sq. units (3).	2+3
38	$\csc\theta - \sin\theta = \dfrac{\cos^2\theta}{\sin\theta}$, $\sec\theta - \cos\theta = \dfrac{\sin^2\theta}{\cos\theta}$ (2); $\tan\theta + \cot\theta = \dfrac{1}{\sin\theta\cos\theta}$ (1); product $= \sin\theta\cos\theta \cdot \dfrac{1}{\sin\theta\cos\theta} = 1$ (2).	5
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 4 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 4$, so $d = 2 + 2\sqrt{3}$ (2); $H = d\tan 60^\circ = 6 + 2\sqrt{3}$; tower $6 + 2\sqrt{3}$ m, distance $2 + 2\sqrt{3}$ m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 63	Twice the area
2	**(c)** $-\frac{7}{25}$	$\cos^2\theta - \sin^2\theta$
3	**(a)** 20 m	Height of the top of the flagstaff $= d\tan 60^\circ = 30$ m; Length $= 30 - 10 = 20$ m
4	**(b)** 14 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 7 = 14$ cm
5	**(d)** 150 mm	Circumference $= \pi d = 66$ mm; diameters $= 4 \times 21 = 84$ mm; Total $= 150$ mm
6	**(b)** 2425.5 cm³	Cylinder $= \pi r^2(2r) = 7276.5$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm^3; Unoccupied $= 2425.5$ cm^3
7	**(b)** 9	$238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$; $x = 9$; then $y = 100 - 82 - x = 9$
8	**(b)** $\frac{1}{6}$	Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
9	**(d)** 9	HCF of the differences from the remainder
10	**(c)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
11	**(d)** 8	Solve for x and y
12	**(b)** 6	$D = 0$ gives $k^2 = 4m$
13	**(a)** 46	$a_n = S_n - S_{n-1}$
14	**(d)** $\frac{12}{5}$	$\frac{1}{2}ab = \frac{1}{2}ch$
15	**(c)** $\frac{16}{3}$	Equate squared distances
16	**(c)** $\frac{24}{25}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
17	**(b)** 2 minutes	Distance covered in 4 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 4 \div 2 = 2$ minutes
18	**(b)** 49 cm	$15x + 4 = 8x + 25$ gives $x = 3$; $QA = 15(3) + 4 = 49$ cm
19	**(a)** 1925 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 35^2$; Area $= 1925$ m^2
20	**(b)** 15 cm	$H = 3h = 3 \times 5$; $H = 15$ cm

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
22	$\cos^2\theta = 1 - \left(\frac{4}{5}\right)^2 = \frac{9}{25}$ (1); value $= -\frac{7}{25}$ (1).	1+1
23	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 6 m; $h(\cot 45^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 9 + 3\sqrt{3}$ m (distances $9 + 3\sqrt{3}$ m and $3 + 3\sqrt{3}$ m) (1).	1+1
24	Area of $\triangle OAP = \frac{1}{2} \times 60 \times 45$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{60 \times 45}{75} = 36$ cm (1); $AB = 2AM = 72$ cm (1).	1+1
25	The 14 ribs divide the circle into 14 equal sectors; Area of the circle $= \frac{22}{7} \times 49^2 = 7546$ cm^2 (1); Area between ribs $= 7546 \div 14 = 539$ cm^2 (1).	1+1
26	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(6)^3 = \pi(8)^2 x$ (1); $x = 4.5$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 18\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 18\sqrt{3}$, so $d = 27$ (1); $H = d\tan 60^\circ = 27\sqrt{3}$; tower $27\sqrt{3}$ m, distance 27 m (1).	1+1+1
28	Area of $\triangle OAP = \frac{1}{2} \times 30 \times 40$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{30 \times 40}{50} = 24$ cm (1); $AB = 2AM = 48$ cm (1).	1+1+1

29	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{50}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 770$ cm^2 (1).	1+1+1
30	Volume of cylinder $= \pi \times 9^2 \times 32$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
31	$\sum f_i = 100$ and $\sum f_i x_i = 7530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7530}{100} = 75.3$ (1); $l = 80$, $f_1 = 23$, $f_0 = 20$, $f_2 = 2$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{3}{24} \times 10 = 81.25$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
32	Required number $=$ HCF$(1250, 9376)$ (1); HCF$(1250, 9376) = 2$ (2).	1+2
	OR: $(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $100 - 16 = 4c$ (1); $c = 21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$XA = XC$ (tangents from X) and $YB = YC$ (tangents from Y) (2); $XY = XC + CY$ since C lies on XY, so $XY = AX + BY$ (1); Perimeter of $\triangle PXY = PX + XY + YP = PX + XA + BY + YP$ (1); $= PA + PB = 2PA$ because $PA = PB$ (1).	2+1+1+1
34	Segment $=$ sector $-$ triangle (2); Sector $= 641.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 529.81$ cm^2 (2); Segment $= 641.67 - 529.81 = 111.85$ cm^2 (1).	2+2+1
35	Slant height of the cone $l = \sqrt{49^2 + 168^2} = 175$ cm (2); TSA $=$ CSA of cone $+$ CSA of hemisphere $= \pi rl + 2\pi r^2$ (2); TSA $= \frac{22}{7} \times 49 \times 175 + 2 \times \frac{22}{7} \times 49^2 = 42042$ cm^2 (1).	2+2+1
36	Median class is 230–240 with $f = 25$, and the cumulative frequency before it is $cf = 20 + x$ (2); $238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$ (2); $x = 9$; then $y = 100 - 82 - x = 9$ (1).	2+2+1
	OR: Total outcomes $= 36$ (2); Favourable outcomes: 4; 8; 15 (2); Probabilities: $\frac{1}{9}$, $\frac{2}{9}$, $\frac{5}{12}$ (1).	2+2+1
37	If $1/\sqrt{2} = r$ were rational then $\sqrt{2} = 1/r$ would be rational - contradiction (3); $\frac{3}{\sqrt{2}} = 3 \times \frac{1}{\sqrt{2}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2
	OR: $\alpha + \beta = 3$, $\alpha\beta = -10$ (1+1); new sum $= -7$ (1); new product $= -30$ (1); polynomial $x^2 + 7x - 30$ (1).	1+1+1+1+1
38	Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $150p + 210q = \frac{41}{4}$, $240p + 120q = 11$ (2); $p = \frac{1}{30}$, $q = \frac{1}{40}$ (2); train 30 km/h, bus 40 km/h (1).	2+2+1
	OR: Boat speed x: $\frac{48}{x-2} + \frac{48}{x+2} = 7$ (1); $48(2x) = 7(x^2 - 4)$ (1); roots of the resulting quadratic: $x = 14$ or negative (1); speed 14 km/h (1).	1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 269.8	$\sum f_i = 100$ and $\sum f_i x_i = 26980$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{26980}{100} = 269.8$
2	**(d)** 45	$\frac{x}{15+x} = 3 \times \frac{15}{15+x}$ gives $x = 3 \times 15$; $x = 45$
3	**(d)** 3	$12 = 2^2 \times 3$; multiply by the primes that occur to an odd power
4	**(c)** $\frac{5}{2}$	$\dfrac{\alpha^2+\beta^2}{\alpha\beta}$
5	**(b)** 12	Intercepts 6 and 4; area $= \frac{1}{2} \times 6 \times 4$
6	**(a)** 2	Substitute the root
7	**(a)** 4	$d = 2p$
8	**(d)** 14	Sides in ratio $a : b$
9	**(b)** 6	$x = 0$: $mc + na = 0$
10	**(b)** $\frac{1}{3}$	$(\sec + \tan)(\sec - \tan) = 1$
11	**(b)** 12 m	Both objects are on the same side, so the distances are subtracted; Distance $= 12$ m
12	**(b)** 120 cm	Half-chord $= \sqrt{68^2 - 32^2} = 60$ cm; Chord $= 2 \times 60 = 120$ cm
13	**(b)** 117 cm	Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm
14	**(d)** 14 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm
15	**(d)** 108	Median class 100–120: $l = 100$, $cf = 44$, $f = 15$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 100 + \frac{50-44}{15} \times 20 = 108$
16	**(d)** 5	Cross-multiplying: $7(7 + x) = 4(16 + x)$; $x = 5$
17	**(d)** $\dfrac{\sqrt{12}}{\sqrt{3}}$	$\sqrt{12}/\sqrt{3} = 2$
18	**(c)** 11	$\frac{c}{a}$ for the monic cubic
19	**(d)** 29	$(x + y)^2 - 2xy = 49 - 20$
20	**(c)** $k \le 3$ and $k \neq 0$	$D = 36 - 12k \ge 0$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$\sum f_i = 83 + p$ and $\sum f_i x_i = 23150 + 250p$; $\frac{23150+250p}{83+p} = 274$ gives $(250 - 274)p = -408$ (1); $p = 17$ (1).	1+1
22	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1
23	Each remainder is short of the divisor by 5 (1); LCM$(12, 18) = 36$, number $= 36 - 5 = 31$ (1).	1+1
24	Sum $= 6$, product $= k$ (1); $k = 6$ (1).	1+1
25	Adding: $8(x + y) = 88$ so $x + y = 11$ (1); subtracting: $x - y = 3$, so $x = 7$, $y = 4$ (1).	1+1
26	$\alpha + \beta = 6$, $\alpha\beta = 5$ (1); $\alpha^2 + \beta^2 = 6^2 - 2(5) = 26$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Need $\dfrac{a+b}{3} = \dfrac{a-b}{1} = \dfrac{14}{7} = 2$ (1); $a + b = 6$, $a - b = 2$ (1); $a = 4$, $b = 2$ (1).	1+1+1
28	$D = 25 - 4k \ge 0$ (1); $k \le \frac{25}{4}$ (1); greatest integer $k = 6$ (1).	1+1+1
29	$d = 4$, $a = 6$ (2); $S_{16} = 576$ (1).	2+1
30	Altitude bisects the base: $h^2 = (6)^2 - 3^2 = 27$ (1+1); $h = 3\sqrt{3}$ cm (1).	1+1+1
31	P divides AB in $1 : 2$: $(3, 2)$ (1.5); Q divides AB in $2 : 1$: $(6, 4)$ (1.5).	1.5+1.5
	OR: Square both sides: $1 + 2\sin\theta\cos\theta = \frac{49}{25}$ (2); $\sin\theta\cos\theta = \frac{12}{25}$ (1).	2+1

32	Distance of the point from the building: $d = 14\cot 45^\circ = 14$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 14\sqrt{3}$ m (1); Length $= 14\sqrt{3} - 14 = 14\sqrt{3} - 14$ m (1).	1+1+1
	OR: $AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 36 - r$, $CD = BC - r = 77 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(36 - r) + (77 - r) = 85$ gives $r = 14$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	AP: $a = 2500$, $d = 150$, $n = 18$ (1); last $= a + (n-1)d = 5050$ (2); total $= S_n = \frac{18}{2}(2500 + 5050) = 67950$ (2).	1+2+2
34	Draw a horizontal line from the top of the shorter pole: right triangle with legs 24 and 18 (2); $x^2 = 576 + 324$ (1); $x = 30$ m (2).	2+1+2
35	Diagonals bisect each other: midpoint of AC is $(1, 2)$ (2); this is also the midpoint of BD (1); $D = (1, 4)$ (2).	2+1+2
36	Divide numerator and denominator by $\cos A$ (1): $\frac{5\tan A + 4}{7\tan A + 3}$ (1); $= \frac{5(\frac{24}{7}) + 4}{7(\frac{24}{7}) + 3} = \frac{148}{189}$ (2); verified with $\sin A = \frac{24}{25}$, $\cos A = \frac{7}{25}$ (1).	1+1+2+1
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 48 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 48$ gives $h = 12\sqrt{3}$ (2); Height $12\sqrt{3}$ m; distances 12 m and 36 m (1).	2+2+1
37	Area of $\triangle OAP = \frac{1}{2} \times 72 \times 21$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (2); $AM = \frac{72 \times 21}{75} = 20.16$ cm (2); $AB = 2AM = 40.32$ cm (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{36}{360} \times \frac{22}{7} \times (21^2 - 14^2)$ (2); Area $= 77$ cm^2 (1).	2+2+1
38	The cylinder has radius r and height $2r$ (2); Cylinder $= \pi r^2(2r) = 196465.5$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 130977$ cm^3 (2); Unoccupied $= 65488.5$ cm^3 (1).	2+2+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 7880$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7880}{100} = 78.8$ (2); $l = 50$, $f_1 = 23$, $f_0 = 7$, $f_2 = 19$, $h = 10$ (2); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{16}{20} \times 10 = 58$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER KEY